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Match-play activity profile in professional soccer players during official games and the relationship between external and internal load

Short title: Physical demands in professional soccer players

Luis Suarez-Arrones^{1,2}, Nacho Torreño^{1,3}, Bernardo Requena^{1,2}, Eduardo Sáez de Villarreal^{1,2}, David Casamichana^{1,4}, José Carlos Barbero-Alvarez⁵ & Diego Munguía-Izquierdo^{1,2}

¹*MasterdeFútbol. Pablo de Olavide University, Sevilla, Spain*

²*Department of Sport and Informatics, Section of Physical Education and Sport. Pablo de Olavide University, Sevilla, Spain*

³*Maccabi Tel Aviv Football Club. Israel.*

⁴*Department of Physical Education and Sport. University of the Basque Country, Victoria-Gasteiz, Spain.*

⁵*Department of Physical Education and Sport. University of Granada, Spain.*

Address for correspondence:

Luis Suarez Arrones, PhD

Faculty of Sport

Pablo de Olavide University

Ctra. de Utrera km 1

41013, Sevilla

Spain

Phone: (+34) 652914248

E-mail: ljsuamor@upo.es

Abstract

Aim: The aim was to quantify for the first time the physical and physiological profile of professional soccer players in official games using GPS and heart rate response. **Methods:** Thirty professional soccer players were investigated during a half in competitive club level matches ($n=348$) using GPS devices. **Results:** The relative total distance was 118.9 ± 10.7 m·min⁻¹ and player's work-to-rest ratio was 2.1:1. Defenders covered the lowest total distance, while Second-Strikers (2ndS) and Wide-Midfielders (W-MD) traveled the greatest total distance. Defenders presented the lowest work-to-rest ratio values. Playing position also impacted on all sprinting performance results, except in average sprint distance and time of sprint. The number of sprints and repeated-sprint sequences recorded by the W-MD and Strikers (S) were significantly greater than any other group. The average Heart Rate (HR) recorded was $87.1\%HR_{max}$ and the relationship between the external and internal load value (effindex) was 1.4 with significant differences in both between playing positions. W-MD recorded a significantly smaller average HR than any other group and Centre-Backs showed a significantly smaller effindex value than any other group. Conversely, W-MD showed a significantly greater effindex value than any other group, except the 2ndS. **Conclusion:** This study has verified a number of statistically significant differences between the different playing positions. Coaches should be focused on the specific physical and physiological requirements of the playing positions to optimize the training prescription in soccer. The relationships between external and internal load measures among position-specific indicates that players with less overall running performance during match-play were the worst in effindex.

Keywords: team sport, time motion, GPS, football

Introduction

Football is a predominantly aerobic sport, interspersed with frequent bouts of high-speed movements ^{1,2}. Understanding the physical demands of match-play is important in order to optimise the training process, since specific protocols can then be designed in accordance with these demands ³.

Over the years, numerous techniques have been used to determine the physical profile of soccer players ³⁻⁵. In this context, the recent introduction of GPS technology for monitoring both training and competitive events provides a reliable and valid measure of the physical profile ^{6,7}. Furthermore, the use of GPS looks set to increase our knowledge by revealing how a soccer player's physical profile evolves with age ⁸, how it varies according to the playing position ^{8,9}, and between competitive versus training contexts ⁴. Due to the differences between video-based analysis system and GPS devices care should be taken when using data from them interchangeably to log the external load on soccer players ^{10,11}.

Although GPS devices are now being used during training by numerous soccer clubs, very little information is available about their application to the study of physical profiles in players during competition. Some studies have used GPS devices to describe the physical profile among young soccer players in different age categories ^{8,9}, semi-professional soccer players ⁴ in friendly matches, or acceleration profile in Australian soccer players ². No published studies have described the physical and physiological profile of European professional soccer players using GPS technology in official games. Accordingly, the aim of this study was to quantify for the first time the physical and physiological profile of elite European professional soccer

players in official games using GPS and heart rate response, quantifying the dose-response of the match stimulus (external and internal load).

Materials and methods

Subjects

Time-motion analysis of running activity and heart rate response was collected in 30 elite European professional soccer players (~14h of training per week). The participants were informed in detail about the experimental procedures and the possible risks and benefits of their participation. The study was approved by the Research Ethics Committee of Pablo de Olavide University and written informed consent was obtained from all participants. Players were informed that they could refuse to participate in the study.

Experimental protocol

An observational design was used to examine the physical demands and exercise intensity in male soccer players during competitive matches using GPS technology and HR response. Thirty elite professional outfield players from the same team were investigated during competitive club level matches in the national league, national cup, and national super cup. Match analyses were performed 4-15 times on each player over a period of two seasons ($n = 348$). At the same time, the team also was participating in the Champions League or UEFA Europa League, depending on season. Tactically, the team used a 4-4-1-1 formation, a variation of 4-4-2 with 1 of the strikers playing as a “second striker”. In order to reduce possible fatigue, tactical changes or the possible influence of the result in the running performance during the second half, only the first half was analysed to describe movement patterns during match-play (45-min each plus additional time). Only time-motion data of

players who participated in the entire half were retained for the subsequent analyses. GPS and heart rate (HR) data were used to provide quantitative information on match-play.

Activity pattern measurements

Players were required to wear a GPS unit (SPI Pro X; GPSports Systems, Canberra, Australia) during the game, which was fitted to the upper back (i.e., between the shoulder blades) of each player using an adjustable neoprene harness. The devices were switched on ~10 minutes before the game and immediately switched off at the end. The data stored includes HR, time, speed, distance, accelerations and impacts (“g” force). GPS data were recorded at 5 Hz and accelerometer data at 100 Hz respectively. The validity and reliability of the GPS system have been previously reported ^{6,12} and used in soccer players ^{2,4,8,9}.

Match running demands analysis

Only the time-motion data for players who participated in the entire first half were retained and only were analysed the minutes of the match-play. Time-motion data from all players were analysed with the software designed to provide objective measures of physical match performance. The players’ activities were coded into five categories and speed thresholds ⁴: walking (0.1-7.0 km·h⁻¹), running at low-speed (7.1-13.0 km·h⁻¹), at medium-speed (13.1-18.0 km·h⁻¹), at high-speed (18.1-21.0 km·h⁻¹), and at sprint (>21.0 km·h⁻¹). The above categories were divided in two further locomotor categories to provide an estimation of player exercise-to-rest ratios; (a) low-speed activity (0.1-7 km·h⁻¹) and (b) moderate-and-high-speed activity (>7.0 km·h⁻¹). Sprint activities were defined as at least 1-s run >21 km·h⁻¹. Repeated-sprint sequence (RSS) was defined as a minimum of two consecutive ≥1-s sprints interspersed with a maximum of 60-s of recovery.

Exercise intensity

Match exercise intensity was quantified by monitoring heart rate (HR) during each game. HR was continuously measured with short-range telemetry (SPI-Pro-X, GPSports, Australia) and was expressed in relation to the individual maximal HR (HR_{max}) obtained throughout an incremental field test¹³ (the highest 5-s average recorded during the test) or during the course of the match if the HR_{max} was higher than the value obtained during the test.

Relationship between external and internal load measures

The performance efficiency (Effindex) for the quantification of the dose-response of the match stimulus was calculated as [mean speed in $m \cdot min^{-1}$ /mean exercise intensity ($\%HR_{max}$)] for every entire first half¹⁴⁻¹⁶. This index integrates mean speed (i.e. external load) with respect to the relative cardiovascular stress (i.e. internal load) during the match-play into a single parameter

¹⁴.

Statistical analysis

All statistical tests were performed using the Social Sciences package (SPSS, 2010, IBM SPSS Statistics 19 Core System User's Guide; SPSS Inc., Chicago, IL). Descriptive statistics were calculated on each variable and Shapiro-Wilk test were used to verify normality. A one-way analysis of variance (ANOVA) was used to determine differences between playing positions. In the event of a significant difference, Bonferroni's post-hoc tests were used to identify any localized effects. Statistical significance was set at $P < 0.05$. Data are presented as means and standard deviations (SD).

Results

Movement Analysis

The relative total distance (RTD) ($\pm$ SD) covered during the half was $118.9 \pm 10.7 \text{ m}\cdot\text{min}^{-1}$. As a percentage of this distance, 32.2% ($38.3 \pm 4.5 \text{ m}\cdot\text{min}^{-1}$) was spent walking, 38.4% ($45.7 \pm 6.7 \text{ m}\cdot\text{min}^{-1}$) running at low-speed, 19.7% ($23.4 \pm 5.8 \text{ m}\cdot\text{min}^{-1}$) at medium-speed, 5.4% ($6.4 \pm 2.1 \text{ m}\cdot\text{min}^{-1}$) at high-speed and 4.2% ($5.0 \pm 2.5 \text{ m}\cdot\text{min}^{-1}$) at sprint. Player's work-to-rest ratio was 2.1:1. Match running profiles are shown in Table 1. Playing position significantly impacted on distance covered during the half (45-min) and on work-to-rest ratio. Defenders covered the lowest RTD. Centre-Backs (CB) covered a $>18 \text{ km}\cdot\text{h}^{-1}$ distance significantly shorter than any other group. Conversely, Second-Strikers (2ndS) and Wide-Midfielders (W-MD) covered the greatest RTD. W-MD covered a $>18 \text{ km}\cdot\text{h}^{-1}$ and $>21 \text{ km}\cdot\text{h}^{-1}$ distance significantly greater than any other group. Defenders presented the lowest work-to-rest ratio values. CB showed a significantly smaller work-to-rest ratio than any other group, whereas Strikers (S) did not show significantly greater values than defenders.

The number of sprints, average sprint distance and maximal sprint distance during the half (45-min) were: 12.7 ± 6.1 sprints, $18.2 \pm 3.4 \text{ m}$, and $35.1 \pm 11.3 \text{ m}$, respectively. Sprint duration was $2.50 \pm 0.5 \text{ s}$ with one sprint being performed on average every $4.1 \pm 2.4 \text{ min}$ and the number of RSS during the half (45-min) was 3.1 ± 2.9 . Playing position significantly impacted on all sprinting performance results, except in average sprint distance and time of sprint. The number of sprints and RSS recorded by the W-MD and S were significantly greater than any other group. The maximal sprint distance and average minutes between sprints recorded by the W-MD were significantly greater and smaller, respectively, than any other group, except the S (Table 2).

Exercise intensity

The average HR recorded during the half was 87.1% HR_{max} with significant differences between playing positions. W-MD recorded a significantly smaller average HR than any other group (Figure 1).

Relationship between external and internal load measures

The effindex value was 1.4 with significant differences between playing positions. CB showed a significantly smaller effindex value than any other group. Conversely, W-MD showed a significantly greater effindex value than any other group, except the 2ndS (Figure 1).

Discussion

The aim of this study was to quantify for the first time the physical and physiological profile of elite professional soccer players in official games using GPS and heart rate response. The main findings of the present study reflected that 1) playing position impacted significantly on relative distance covered, work-to-rest ratio, sprinting performance results and average HR, and 2) relationships between external and internal load measures among position-specific players indicated that those with less overall running performance during match-play showed the worst results in effindex.

The overall running-patterns ($119 \text{ m} \cdot \text{min}^{-1}$) and sprinting performance results in the present study were higher than those previously reported in semi-professional soccer players ⁴ ($113 \text{ m} \cdot \text{min}^{-1}$), Australian soccer players ¹⁷ ($104 \text{ m} \cdot \text{min}^{-1}$) and youth soccer players ⁸ (ranging from 93.5 to $108.8 \text{ m} \cdot \text{min}^{-1}$), indicating that elite European professional soccer official games are played at higher relative running demands. The work-to-rest ratio values (2.1:1) in the present

study were smaller than those reported in semi-professional soccer players ⁴ (2.4:1). These data suggest that higher playing standards in soccer might be associated with overall increased running demands and sprinting performance. However, comparisons should be made with caution because different GPS systems [10-Hz ⁴, 5-Hz ¹⁷ and 1-Hz ⁸] and time-motion analysis methods were employed (0.1-7 km·h⁻¹ for rest in the present study vs. 0.1-4 km·h⁻¹ in semi-professional soccer players), and previous studies revealed rather large between-system differences in the determination of distance covered ^{10,11}. Nevertheless, the present study reflects the specific demands during match-play (official games) in professional soccer players using GPS devices, and can be used to develop position-specific training tasks based on this tactical formation. Also, the work-to-rest ratio obtained can also help coaches to plan specific intermittent exercise training protocols.

Defenders presented the lowest values of RTD and work-to-rest ratio. CB showed the lowest values of distance at high-speed, work-to-rest ratio and effindex, while 2ndS and W-MD presented the highest values of RTD and effindex. W-MD was the playing position with significantly lowest average HR, highest high-speed running and sprinting values. Probably, the wide differences between CB and W-MD observed in the current study could be related to the tactical roles and training adaptations. W-MD have a linking role in the team and need to complete fast movement away from defending players, take advantage on goal scoring opportunities ¹⁸ and also back to defend when their team lose the ball. While the average sprint distance was similar for all specific positions (~ 18 m), with this tactical formation and game philosophy W-MD presented the highest values in maximal sprint distance and the highest number of RSS (41 m and 6 RSS during the half, respectively), excluding the S who reported similar values (39 m and 5 RSS during the half, respectively). Specific positions showed

maximal sprint distance from 32 to 41 m with averages of peaks of speed from 28 to 31 km·h⁻¹. A previous study reported as a rule, speed sessions typically conducted by fitness trainers include short sprints (< 20 m) ¹⁹. Therefore, this information should be used for the prescription of drills that replicate game repeated-sprint sequences, sprint distances and peaks of speed, in order to prescribe specific trainings adapted to individual game demands; and the assessment of running speed in developing players should therefore include assessment of both acceleration and distances ~ 40 m ¹⁹.

Our results confirm data on position-specific running patterns reported in adult soccer players using video-analysis ^{3,20} or in youth soccer players ²¹. Therefore, this information reflects that players have specific demands during match-play and can be used to develop position-specific training strategies. Moreover, the effindex expressed how efficiently one can run with a given cardiovascular stress ¹⁴. Accordingly, the relationships between external and internal load measures among position-specific in the present study indicates that players with less overall running performance during match-play were the worst in effindex.

Conclusion

The assessment of the external (i.e., running demands using GPS devices) and internal (i.e., HR responses) load imposed during the competition is the first step preceding the design of position-specific conditioning training tasks controlled with this technology. This study has quantified physical and physiological demands of professional soccer players with GPS and HR response and has verified a number of statistically significant differences between the different playing positions. Coaches should be focused on the specific physical and

physiological requirements of the playing positions to optimize the training prescription in soccer. Future studies in professional soccer players adding physical performance assessments and using relative intensity zones during official games are needed.

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Table 1. Match running profile (only first half) in professional soccer players ($n = 348$). Data are mean $\pm$ SD.

Table 2. Sprinting performance results obtained during matches (only first half) in professional soccer players ($n = 348$). Data are mean $\pm$ SD.

Figure 1. Average Heart Rate expressed in relation to the individual maximal HR (HR_{max}) and Effindex during the half in professional soccer players ($n = 348$). Data are mean $\pm$ SD.

CB: Centre Backs; **FB:** Full Backs; **MD:** Midfielders; **W-MD:** Wide Midfielders; **2ndS:** Second Strikers; **S:** Strikers.

a: Significant difference vs. CB; **b:** Significant difference vs. FB; **c:** Significant difference vs. MD; **d:** Significant difference vs. W-MD; **e:** Significant difference vs. 2ndS; **f:** Significant difference vs. S; *: Significantly smaller than any other group.

Table 1. Match running profile (only first half) in professional soccer players (n = 348). Data are mean ± SD.

<i>Variables</i>	CB (<i>n</i> = 53)	FB (<i>n</i> = 57)	MD (<i>n</i> = 79)	W-MD (<i>n</i> = 68)	2 nd S (<i>n</i> = 44)	S (<i>n</i> = 47)
Relative total distance (m·min ⁻¹)	103.7 ± 7.3 [*]	112.8 ± 6.7 ^{c,d,e,f}	122.6 ± 6.8 ^e	125.6 ± 6.9	127.7 ± 6.8	119.1 ± 8.7 ^{d,e}
RD: 0.1 - 7.0 km·h ⁻¹ (m·min ⁻¹)	40.7 ± 3.3	39.3 ± 4.2	36.8 ± 2.9 ^{a,b,f}	36.3 ± 3.1 ^{a,b,f}	36.8 ± 4.2 ^{a,b,f}	41.0 ± 5.7
RD: 7.1 - 13.0 km·h ⁻¹ (m·min ⁻¹)	38.2 ± 4.7 ^{b,c,d,e,f}	43.6 ± 4.9 ^{c,d,e}	50.0 ± 5.8	47.6 ± 4.4	48.2 ± 5.3	43.9 ± 7.8 ^{c,d,e}
RD:13.1 - 18.0 km·h ⁻¹ (m·min ⁻¹)	17.1 ± 3.2 ^{b,d,e,f}	19.8 ± 3.7 ^{c,d,e}	26.2 ± 4.8 ^e	25.8 ± 3.9 ^e	29.7 ± 3.5	20.9 ± 4.6 ^{c,d,e}
RD:18.1 - 21.0 km·h ⁻¹ (m·min ⁻¹)	4.4 ± 1.4 ^{b,d,e,f}	5.4 ± 1.6 ^{d,e,f}	6.0 ± 1.9 ^{d,e}	8.1 ± 1.6	7.9 ± 1.8	6.5 ± 1.4 ^{d,e}
RD: >21.0 km·h ⁻¹ (m·min ⁻¹)	3.3 ± 1.9 ^{b,d,e,f}	4.6 ± 2.2 ^{d,f}	3.4 ± 1.7 ^{b,d,e,f}	7.6 ± 2.1 ⁺	5.0 ± 1.8 ^{d,f}	6.6 ± 1.8 ^d
RD: > 13 km·h ⁻¹ (m·min ⁻¹)	24.7 ± 5.2 ^{b,c,d,f}	29.9 ± 6.3 ^{c,d,e,f}	35.6 ± 6.8 ^{d,e}	41.6 ± 6.3	42.6 ± 5.2	34.0 ± 6.0 ^{d,e}
RD: > 18 km·h ⁻¹ (m·min ⁻¹)	7.6 ± 2.9 [*]	10.0 ± 3.4 ^{d,e,f}	9.4 ± 3.2 ^{d,e,f}	15.7 ± 3.2 ⁺	12.9 ± 3.1 ^d	13.1 ± 2.7 ^d
Work-to-rest ratio (						
RD>7.0 km·h ⁻¹ / RD≤7.0 km·h ⁻¹)	1.56 ± 0.25 [*]	1.90 ± 0.36 ^{c,d,e}	2.34 ± 0.33 ^f	2.15 ± 0.50 ^f	2.51 ± 0.46 ^f	1.95 ± 0.49

CB: Centre Backs; **FB:** Full Backs; **MD:** Midfielders; **W-MD:** Wide Midfielders; **2ndS:** Second Strikers; **S:** Strikers. **RD:** Relative distance.

a: Significant difference vs. CB; **b:** Significant difference vs. FB; **c:** Significant difference vs. MD; **d:** Significant difference vs. W-MD; **e:** Significant difference vs. 2ndS; **f:** Significant difference vs. S; *****: Significantly smaller than any other group; **+**: Significantly greater than any other group.

Table 2. Sprinting performance results obtained during matches (only first half) in professional soccer players ($n = 348$). Data are mean $\pm$ SD.

<i>Variables</i>	CB ($n = 53$)	FB ($n = 57$)	MD ($n = 79$)	W-MD ($n = 68$)	2 nd S ($n = 44$)	S ($n = 47$)
Number of Sprints	7.9 $\pm$ 3.9 ^{b,d,e,f}	11.3 $\pm$ 4.8 ^{d,f}	8.9 $\pm$ 4.4 ^{b,d,e,f}	18.7 $\pm$ 5.2	13.2 $\pm$ 4.5 ^{d,f}	16.6 $\pm$ 4.7
Average time of sprint (s)	2.5 $\pm$ 0.7	2.5 $\pm$ 0.5	2.4 $\pm$ 0.5	2.6 $\pm$ 0.3	2.4 $\pm$ 0.5	2.5 $\pm$ 0.3
Average minutes between sprint	5.7 $\pm$ 2.8 ^{b,d,e,f}	4.4 $\pm$ 2.7 ^{d,f}	5.4 $\pm$ 2.7 ^{d,e,f}	2.5 $\pm$ 0.9	3.7 $\pm$ 1.5 ^d	2.8 $\pm$ 0.8
Average Sprint Distance	18.4 $\pm$ 4.3	18.7 $\pm$ 3.8	17.6 $\pm$ 3.4	18.6 $\pm$ 2.7	17.6 $\pm$ 3.4	18.2 $\pm$ 2.7
Maximal Sprint Distance	31.8 $\pm$ 12.7 ^{d,f}	34.6 $\pm$ 10.2 ^d	31.1 $\pm$ 10.6 ^{d,f}	41.0 $\pm$ 10.0	34.0 $\pm$ 10.2 ^d	38.8 $\pm$ 10.8
Maximal Speed	27.9 $\pm$ 2.7 ^{b,d,f}	29.3 $\pm$ 1.8	27.7 $\pm$ 2.4 ^{b,d,f}	31.0 $\pm$ 2.8	28.5 $\pm$ 2.1 ^{d,f}	30.1 $\pm$ 2.3
RSS	1.0 $\pm$ 1.7 ^{b,d,e,f}	2.5 $\pm$ 2.2 ^{d,f}	1.7 $\pm$ 1.7 ^{d,e,f}	5.8 $\pm$ 3.2	3.2 $\pm$ 2.5 ^{d,f}	4.8 $\pm$ 2.8

CB: Centre Backs; **FB:** Full Backs; **MD:** Midfielders; **W-MD:** Wide Midfielders; **2ndS:** Second Strikers; **S:** Strikers. **RSS:** Repeated-sprint sequence.

a: Significant difference vs. CB; **b:** Significant difference vs. FB; **c:** Significant difference vs. MD; **d:** Significant difference vs. W-MD; **e:** Significant difference vs. 2ndS; **f:** Significant difference vs. S.

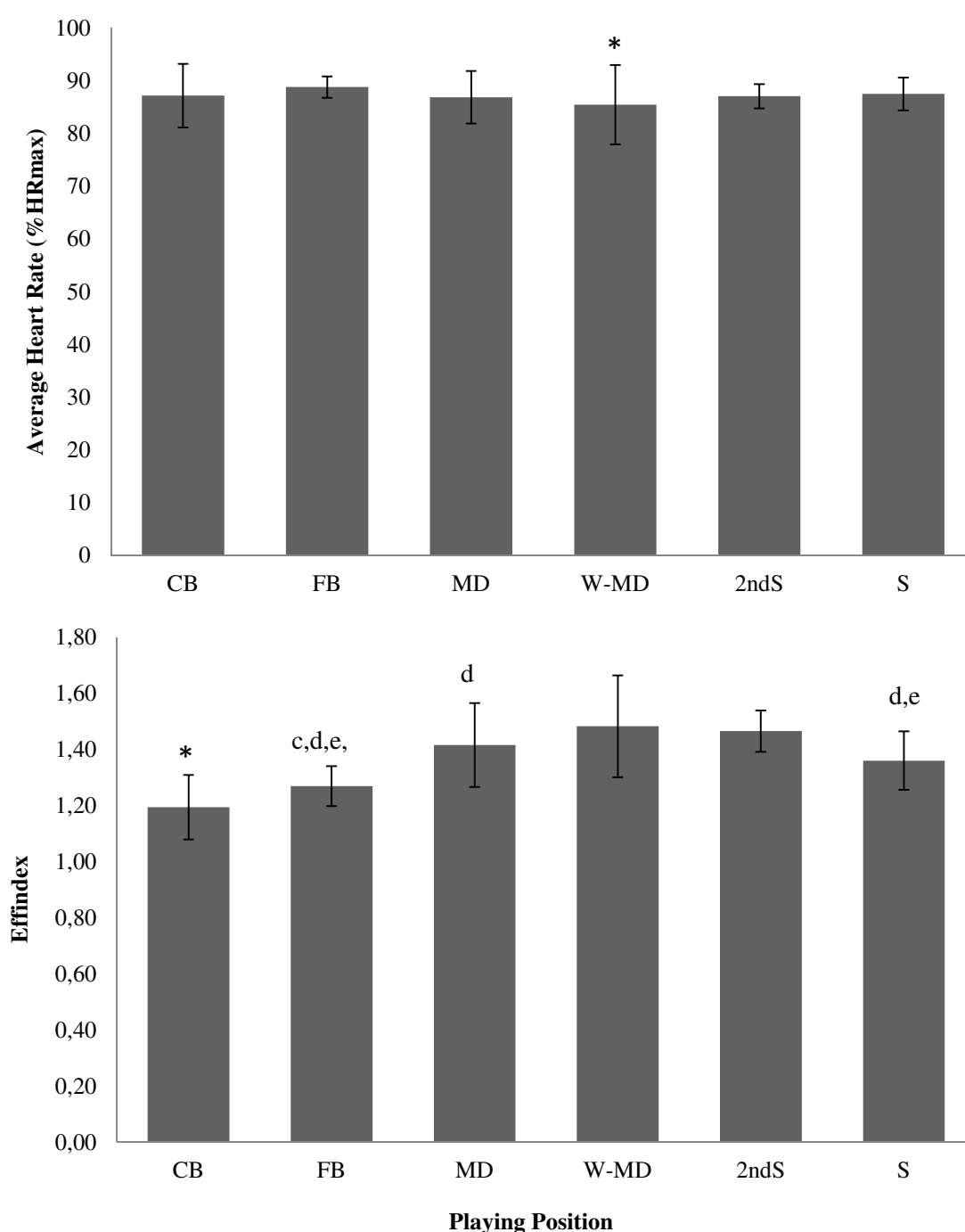


Figure 1. Average Heart Rate expressed in relation to the individual maximal HR (HR_{max}) and Effindex during the half in professional soccer players ($n = 348$). Data are mean $\pm$ SD.

CB: Centre Backs; **FB:** Full Backs; **MD:** Midfielders; **W-MD:** Wide Midfielders; **2ndS:** Second Strikers; **S:** Strikers.

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